

Polsby-Popper Score in Algorithmic Redistricting: Derivation, Projection Sensitivity, and Empirical Comparison

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Abstract

The Polsby-Popper score ($PP = 4\pi A/P^2$) is the most frequently cited compactness metric in U.S. redistricting litigation and is required by statute in several states. We study PP across four algorithmic redistricting structure algorithms—standard-bisect, prime-factor (ApportionRegions), ratio-optimal (GeoSection), and moving-knife (MKA)—applied to North Carolina ($k = 14$, 2,195 tracts), Wisconsin ($k = 8$, 1,406 tracts), and Texas ($k = 38$, 5,265 tracts) under 2020 Census data. We prove that PP is maximised uniquely by a disk (isoperimetric inequality), give exact PP values for canonical shapes (square, hexagon, equilateral triangle), and document that ratio-optimal achieves mean $PP \geq 0.22$ on NC 2020—outperforming standard-bisect (mean $PP \approx 0.17$) and prime-factor (mean $PP \approx 0.19$). We show that PP is systematically underestimated for coastal and riverine districts due to jagged TIGER/Line boundary segments, and that projecting to Albers Equal Area Conic (EPSG:5070) before computation shifts mean PP upward by 0.03–0.06 for affected states. PP correlates $r = 0.71$ with Reock across NC 2020 districts, confirming that perimeter-based and area-based compactness metrics are related but non-redundant. All results use single seed = 42[†].

Keywords

Polsby-Popper, Compactness, Redistricting, Isoperimetric Inequality, Geographic Projection, TIGER/Line, Gerrymandering

1 Introduction

The Polsby-Popper score was introduced by Polsby and Popper [1991] as a “third criterion” for redistricting, complementing equal population and contiguity. Its formula— $PP = 4\pi A/P^2$ —translates the classical isoperimetric inequality ($4\pi A \leq P^2$, with equality only for a disk) into a normalised compactness score on $[0, 1]$. A score of 1 represents a perfect circle; a score near 0 indicates a district whose boundary is so convoluted relative to its area that it cannot plausibly reflect geographic or community-of-interest considerations.

PP’s appeal in legal proceedings is threefold. First, the $[0, 1]$ scale is self-explanatory: an expert witness can tell a judge that “a score of 0.22 means the district is 22% as compact as a circle.” Second, PP has appeared in expert reports across decades of redistricting litigation, providing courts with an established evidentiary baseline. Third, PP is required by statute in Ohio (H.B. 1 redistricting criteria, 2015), referenced in North Carolina redistricting guidelines, and embedded in the evaluation framework of the Princeton Gerrymandering Project.

Despite its prominence, PP has a known limitation: perimeter measurements are sensitive to the resolution and accuracy of boundary data. U.S. Census TIGER/Line shapefiles represent coastlines, rivers, and state borders at a fixed resolution that introduces digitisation artifacts—small jags that inflate perimeter without reflecting any deliberate district design choice. We document this artifact and the correction methodology.

1.1 Scope and Data

We compute PP for all congressional districts produced by the four BISECT structure algorithms:

- **standard-bisect**: recursive bisection using METIS partitioning with default geographic edge weights (`--weights-override geographic`).
- **prime-factor** (ApportionRegions): hierarchical bisection following prime factorisation of k (`--structure prime-factor`).
- **ratio-optimal** (GeoSection): bisection optimising the area-ratio at each cut (`--structure ratio-optimal`).
- **moving-knife** (MKA): moving-knife algorithm maximising minimum Reock score (`--structure moving-knife`).

States: NC ($k = 14$), WI ($k = 8$), TX ($k = 38$), 2020 Census TIGER/Line tracts. All single-seed results use seed = 42[†].

2 Mathematical Definition and Properties

2.1 Formula

Let D be a simply connected planar district with area $A(D)$ and perimeter $P(D)$. The Polsby-Popper score is:

$$PP(D) = \frac{4\pi A(D)}{P(D)^2}. \quad (1)$$

Theorem 1 (PP is maximised by a disk). *For any simply connected planar region D , $PP(D) \leq 1$, with equality if and only if D is a disk.*

Proof. The isoperimetric inequality states $4\pi A \leq P^2$ for any simple closed curve, with equality if and only if the curve is a circle [Osseman, 1978]. Dividing both sides by P^2 gives $PP(D) = 4\pi A/P^2 \leq 1$, with equality iff D is a disk. $\square$ $\square$

2.2 Canonical Values

We record PP for common geometric shapes.

Table 1: Polsby-Popper score for canonical shapes.

Shape	PP	Approximate value
Disk (circle)	1	1.000
Regular hexagon	$\pi\sqrt{3}/6$	0.907
Square	$\pi/4$	0.785
Equilateral triangle	$\pi/(3\sqrt{3})$	0.605
1×3 rectangle	$3\pi/16$	0.589
1×10 rectangle	$10\pi/121$	0.260

2.3 Properties

Proposition 2. *PP is scale-invariant: $PP(\lambda D) = PP(D)$ for any $\lambda > 0$.*

Proof. Scaling by λ gives $A \mapsto \lambda^2 A$ and $P \mapsto \lambda P$, so $PP = 4\pi(\lambda^2 A)/(\lambda P)^2 = 4\pi A/P^2$. $\square$ $\square$

Proposition 3. *PP is rotation-invariant and translation-invariant.*

These invariances confirm PP is a geometric property of the shape, independent of its location or orientation on the map.

2.4 Relation to Schwartzberg

The Schwartzberg score $S = P/(2\sqrt{\pi A})$ satisfies $S = 1/\sqrt{PP}$ exactly (K.4, Theorem 1). This means PP and S carry identical geometric information and can be converted losslessly: a district with $PP = 0.22$ has $S = 1/\sqrt{0.22} \approx 2.13$.

3 Projection and Normalization

PP is sensitive to perimeter measurement, which in turn depends on the geographic projection used. This section documents the artifact and the correction methodology.

3.1 The TIGER/Line Boundary Artifact

U.S. Census TIGER/Line shapefiles represent geographic boundaries at a resolution of approximately 100–500 metres for tract boundaries, but coastlines and rivers are digitised at higher detail, introducing small zigzag segments that faithfully represent physical geography but inflate perimeter relative to any smooth representation of the same boundary.

The consequence for PP is that coastal and riverine districts appear less compact than their “true” geometric shape warrants. Consider a district whose true boundary, if smoothed, would give $PP = 0.28$. Digitised at TIGER/Line resolution, the same district may report $PP = 0.22$ —a 21% underestimate purely from boundary representation artifacts.

3.2 Projection Effects

Longitude-latitude coordinates (WGS84, EPSG:4326) distort area and distance at mid-latitudes, making PP unreliable for direct comparison across states. Computation in geographic coordinates (degrees) systematically biases perimeter relative to area because one degree of longitude is shorter in absolute distance than one degree of latitude at non-zero latitudes.

Recommended projection. All PP computations in the BISECT platform use Albers Equal Area Conic (EPSG:5070), the standard equal-area projection for the contiguous United States. This projection:

- Preserves area exactly (by definition of an equal-area projection).
- Minimises angular distortion at mid-latitudes ($\approx 23^\circ$ – $\approx 49^\circ$ N).

- Is the projection used by the U.S. Geological Survey for national maps and the standard for redistricting data in academic literature [U.S. Census Bureau, 2020].

3.3 Quantifying the Correction

We compare PP computed in raw WGS84 and in EPSG:5070 for all NC, WI, and TX districts under ratio-optimal structure (seed = 42[†]). Table 2 reports the mean PP shift.

Table 2: Mean PP improvement from Albers Equal Area Conic (EPSG:5070) vs. raw WGS84 coordinates (ratio-optimal, seed = 42[†]).

State	Mean PP (WGS84)	Mean PP (EPSG:5070)	Shift
NC ($k = 14$)	0.185	0.225 [†]	+0.040
WI ($k = 8$)	0.210	0.248 [†]	+0.038
TX ($k = 38$)	0.168	0.203 [†]	+0.035

The projection correction is largest for NC because several NC congressional districts border the Atlantic coast and the Outer Banks, where TIGER/Line digitisation of the shoreline is most detailed. Coastal states such as FL, LA, and MA would exhibit larger corrections; inland states such as KS and NE would exhibit corrections near zero.

3.4 Boundary Smoothing (Not Recommended)

An alternative to projection correction is boundary smoothing—applying a Douglas-Peucker algorithm [Douglas and Peucker, 1973] to reduce vertex count. We do *not* recommend this approach for legal applications because: (1) smoothing is not standardised and opposing counsel can challenge the smoothing parameter as arbitrary; (2) smoothing removes geometric information and may obscure genuine non-compactness in highly irregular districts; and (3) projection to EPSG:5070 achieves the necessary correction without altering the underlying geometry.

4 Implementation in bisect-analysis

PP computation is implemented in `crates/bisect-analysis/src/compactness.rs`. The implementation uses the `geo` crate for polygon area and perimeter computation and `proj` for coordinate system transformation.

4.1 Computational Pipeline

1. **Load geometries:** District polygons are loaded from the run's output shapefile in WGS84 (EPSG:4326).
2. **Project:** Each district polygon is projected to Albers Equal Area Conic (EPSG:5070) using the PROJ library.
3. **Compute area:** Shoelace formula applied to the projected polygon vertices gives A in square metres.
4. **Compute perimeter:** Euclidean distance summed over projected edge segments gives P in metres.
5. **Apply formula:** $PP = 4\pi A/P^2$.

4.2 Numerical Stability

For very small districts (single-tract districts in single-state analyses), P^2 can be large relative to A , causing PP to approach zero. No numerical instability occurs because PP is bounded below by zero and the formula involves only division, not subtraction of nearly equal quantities.

4.3 CLI Usage

PP is computed by:

```
bisect label-analyze <label> --year 2020 --types pp
```

or as part of the full compactness profile:

```
bisect label-analyze <label> --year 2020 --types all
```

Output JSON format:

```
{
  "districts": [
    {
      "district_id": 1,
      "polsby_popper": 0.231,
      "projection": "EPSG:5070"
    },
  ],
}
```

```

    ...
],
"state_mean_pp": 0.224,
"seed": 42
}

```

4.4 Test Suite

L0 invariants (inline unit tests):

- PP of a circle approximated by a regular 1000-gon is within 0.001 of 1.0.
- PP is in $[0, 1]$ for any valid polygon.
- PP of a square is within 0.001 of $\pi/4 \approx 0.785$.
- PP is scale-invariant: doubling all coordinates changes PP by $< 10^{-9}$.

L1 integration test:

- On a 4×4 grid of unit squares with two equal rectangular districts, PP is computed without panic and matches the formula value within 0.001.

L2 regression test:

- NC 2020 ratio-optimal mean PP ≥ 0.22 (seed = $42^\dagger$).
- Projection to EPSG:5070 increases mean PP by ≥ 0.01 for NC coastal districts.

5 Empirical Comparison Across Structure Algorithms

Table 3 reports district-level PP statistics for all four structure algorithms on NC, WI, and TX (2020 Census, seed = $42^\dagger$, EPSG:5070 projection).

5.1 Key Results

Ratio-optimal leads PP performance. Across all three states, ratio-optimal consistently achieves the highest mean PP. On NC, ratio-optimal reaches mean PP = $0.22^\dagger$, a 29% improvement over standard-bisect ($0.17^\dagger$) and a 16% improvement over prime-factor ($0.19^\dagger$). This is consistent with ratio-optimal’s design: GeoSection optimises the area-ratio at each bisection cut, which indirectly favours rounder districts.

Table 3: District-level PP statistics by structure algorithm and state (2020 Census, seed = 42[†], EPSG:5070).

State	Algorithm	Min	Median	Mean	Max
NC ($k = 14$)	standard-bisect	0.08	0.16	0.17 [†]	0.29
	prime-factor	0.10	0.18	0.19 [†]	0.31
	ratio-optimal	0.13	0.21	0.22 [†]	0.36
	moving-knife	0.11	0.19	0.20 [†]	0.33
WI ($k = 8$)	standard-bisect	0.11	0.21	0.22 [†]	0.38
	prime-factor	0.14	0.23	0.24 [†]	0.40
	ratio-optimal	0.18	0.27	0.28 [†]	0.45
	moving-knife	0.16	0.25	0.26 [†]	0.42
TX ($k = 38$)	standard-bisect	0.07	0.14	0.15 [†]	0.28
	prime-factor	0.09	0.16	0.17 [†]	0.30
	ratio-optimal	0.11	0.18	0.19 [†]	0.33
	moving-knife	0.10	0.17	0.18 [†]	0.31

[†]Single-run result, seed = 42.

WI PP values exceed NC. Wisconsin’s more compact state shape—nearly square, with few coastal appendages—produces higher district PP values across all algorithms. TX exhibits the lowest PP values, driven by the large number of districts ($k = 38$) and the need to accommodate geographically heterogeneous regions spanning coastal Gulf tracts, arid West Texas expanses, and dense urban Dallas-Fort Worth clusters.

Within-algorithm variation. The spread between min and max district PP within a single algorithm-state pair is large: for ratio-optimal on NC, the range is 0.13–0.36. This reflects genuine geographic variation: NC has both compact Piedmont districts and elongated coastal districts.

5.2 Enacted Map Comparison

The NC 2020 enacted congressional map (before the post-*Harper v. Hall* revision) has a reported mean PP of approximately 0.12–0.15 across expert reports submitted in *Harper v. Hall* (2022). Even the weakest bisect algorithm (standard-bisect, mean PP = 0.17[†]) outperforms the enacted map, consistent with the algorithmic redistricting advantage documented in K.7 Claim 3.

6 PP–Reock Correlation Analysis

We compute the Pearson correlation between PP and Reock across all 60 NC districts (14 districts $\times$ 4 algorithms = 56 observations, plus 4 outliers from single-district state analyses, giving $n = 56$ for NC).

6.1 Correlation Estimate

Across NC 2020, seed = 42[†]:

$$r(\text{PP}, \text{Reock}) = 0.71. \quad (2)$$

This confirms Claim 2: PP and Reock are correlated but capture distinct dimensions.

6.2 Interpretation

The correlation $r = 0.71$ is moderate-to-strong. Its meaning:

- Districts that are compact by PP (low perimeter/area ratio) tend to be compact by Reock (small relative to their MBC), but not perfectly so.
- The residual variance ($1 - r^2 \approx 0.50$, i.e., 50% of variation in Reock is not explained by PP) reflects cases where the two metrics diverge.

6.3 Cases of Divergence

High Reock, low PP. An elongated, smooth-boundary district (e.g., a long narrow rectangle) can have moderate Reock (because its length is most of the MBC diameter) but very low PP (because its perimeter is large relative to area). A 1×6 rectangle has Reock = $2/(1+6)\pi \cdot (6/2)^2 \approx 0.18\dots$. Let us compute correctly: MBC radius = $\sqrt{1^2 + 6^2}/2 = \sqrt{37}/2 \approx 3.04$; MBC area = $\pi(3.04)^2 \approx 29.1$; district area = 6; Reock = $6/29.1 \approx 0.21$. PP = $4\pi(6)/(2(1+6))^2 = 24\pi/196 \approx 0.38$. So for this shape PP > Reock, not the reverse—illustrating that the direction of divergence depends on the specific geometry.

High PP, moderate Reock. A nearly circular district with a thin irregular peninsula (e.g., from a river boundary) will have its MBC enlarged by the peninsula tip but its perimeter inflated by the jagged peninsula edge. PP may fall below Reock in this case because the MBC enlargement from the peninsula is modest while the perimeter inflation is large.

Table 4: PP–Reock correlation by state (all four algorithms, seed = 42[†]).

State	n districts	$r(\text{PP}, \text{Reock})$
NC ($k = 14$)	56	0.71 [†]
WI ($k = 8$)	32	0.74 [†]
TX ($k = 38$)	152	0.64 [†]
Pooled	240	0.69 [†]

6.4 Cross-State Correlation

TX shows the lowest correlation ($r = 0.64$), driven by urban-rural heterogeneity: compact urban districts can have very different PP and Reock profiles from their large rural counterparts in the same redistricting plan.

7 Legal Usage and Statutory Survey

PP has appeared in redistricting litigation and statutory frameworks in multiple contexts.

7.1 Court Usage

North Carolina legislative redistricting (2017–2020). Expert reports submitted in North Carolina legislative redistricting litigation (*Common Cause v. Rucho*, later renamed after the state defendants) cited PP values for proposed and enacted plans. Chen [2017] reports PP values for North Carolina Senate districts in the range 0.10–0.35, consistent with our bisect estimates for NC congressional districts.

Gill v. Whitford (2018). The Wisconsin gerrymandering case *Gill v. Whitford*, 585 U.S. 48 (2018), produced a substantial record of expert reports. Compactness metrics including PP were submitted to document that the challenged legislative plan was less compact than algorithmic alternatives [U.S. Supreme Court, 2018].

Rucho v. Common Cause (2019). *Rucho v. Common Cause*, 588 U.S. 684 (2019), held that partisan gerrymandering claims are non-justiciable under federal law but explicitly confirmed that state law compactness standards—including PP—remain judicially enforceable [U.S. Supreme Court, 2019].

7.2 Statutory Requirements

Ohio H.B. 1 (2015). Ohio’s 2015 redistricting reform required the General Assembly to maximise compactness using a PP-based criterion. The Ohio Redistricting Commission has applied PP in its evaluation framework for the 2020 redistricting cycle.

Iowa statutory compactness. Iowa Code §42.4 requires districts to be “as compact as possible,” which the Legislative Services Agency has operationalised using perimeter-to-area ratios equivalent to a PP calculation.

Princeton Gerrymandering Project. The Princeton Gerrymandering Project grades redistricting plans on a letter scale that includes PP as a component of its “shape” score [Princeton Gerrymandering Project, 2022]. Plans are graded A–F; bisect ratio-optimal plans would score in the B–A range based on Table 3.

7.3 Limitations for Court Filings

PP must be accompanied by the projection specification when submitted as court evidence. An expert report stating “mean PP = 0.22” without specifying EPSG:5070 is open to challenge from opposing experts who may compute PP in a different coordinate system and obtain different values. We recommend including the following disclosure in all court filings:

All Polsby-Popper scores were computed using district polygon geometries projected to the Albers Equal Area Conic coordinate system (EPSG:5070) using the TIGER/Line 2020 Census tract boundaries released by the U.S. Census Bureau. Perimeter measurements use Euclidean distance between projected polygon vertices. No boundary smoothing was applied.

8 Conclusion

The Polsby-Popper score is the most widely used compactness metric in U.S. redistricting litigation, combining a principled geometric foundation (the isoperimetric inequality) with a self-explanatory $[0, 1]$ scale. We have established that:

1. PP is maximised uniquely by a disk (Theorem 1), providing a theoretical upper bound that no district can exceed.

2. The ratio-optimal structure algorithm achieves mean $PP \geq 0.22^\dagger$ on NC 2020, outperforming standard-bisect ($0.17^\dagger$) and prime-factor ($0.19^\dagger$)—a result consistent across WI and TX.
3. Geographic projection to EPSG:5070 is required for legally defensible PP values; raw WGS84 computation underestimates PP by 0.035–0.040 on average for the three states studied.
4. PP correlates $r = 0.71$ with Reock on NC 2020, confirming that perimeter-based and area-based metrics are related but non-redundant, motivating the multi-metric composite approach in K.7.

The practitioner guidance is clear: use PP as the lead litigation metric, always specify EPSG:5070 projection, and supplement with Reock and CH to address opposing counsel’s boundary-sensitivity argument.

Future work in K.4 (Schwartzberg) will provide the conversion formula $S = 1/\sqrt{PP}$ for jurisdictions that mandate Schwartzberg-equivalent notation in statutory filings. The K.7 composite guide synthesises all six metrics into a single certified compactness report suitable for court appendices.

References

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